DP Biology – D4.3 (HL extension)

Timing, life cycles and evolution

- What is driving climate change?
- How does climate change affect ecosystems?

HL statements D4.3.9–12. Textbook pp. 834–837.
Image: a tawny owl. Akumiszcza, CC BY 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D4.3.9–10	Explain what controls the timing of biological events, and how climate change disrupts it
2	D4.3.11	Explain why warming gives insects more life cycles in a year
3	D4.3.12	Explain evolution in response to climate change

Phenology

- **Phenology** is the study of the timing of biological events
- Examples: flowering, budburst, bud set, bird migration and nesting
- Long records come from scientists and citizen volunteers
- Many records show spring events arriving earlier



Flowering time is a phenological event. Phil aplosh, CC BY 2.0.

Do it

Choose a tree at school. Record the date of budburst each year and share it with a phenology network.

Two main cues

D4.3.9

	Photoperiod (day length)	Temperature
Changes with climate?	No: set by date and latitude	Yes: warming brings changes earlier
Example	Bud set in autumn; timing of bird migration	Budburst in spring; insect development

Key idea

Species in the same ecosystem may use different cues. If one follows temperature and another follows day length, climate change shifts one but not the other, and events that used to happen together drift apart.

Out of step: reindeer and Arctic plants

D4.3.10



Arctic plants, Alaska. BLM Alaska, public domain.

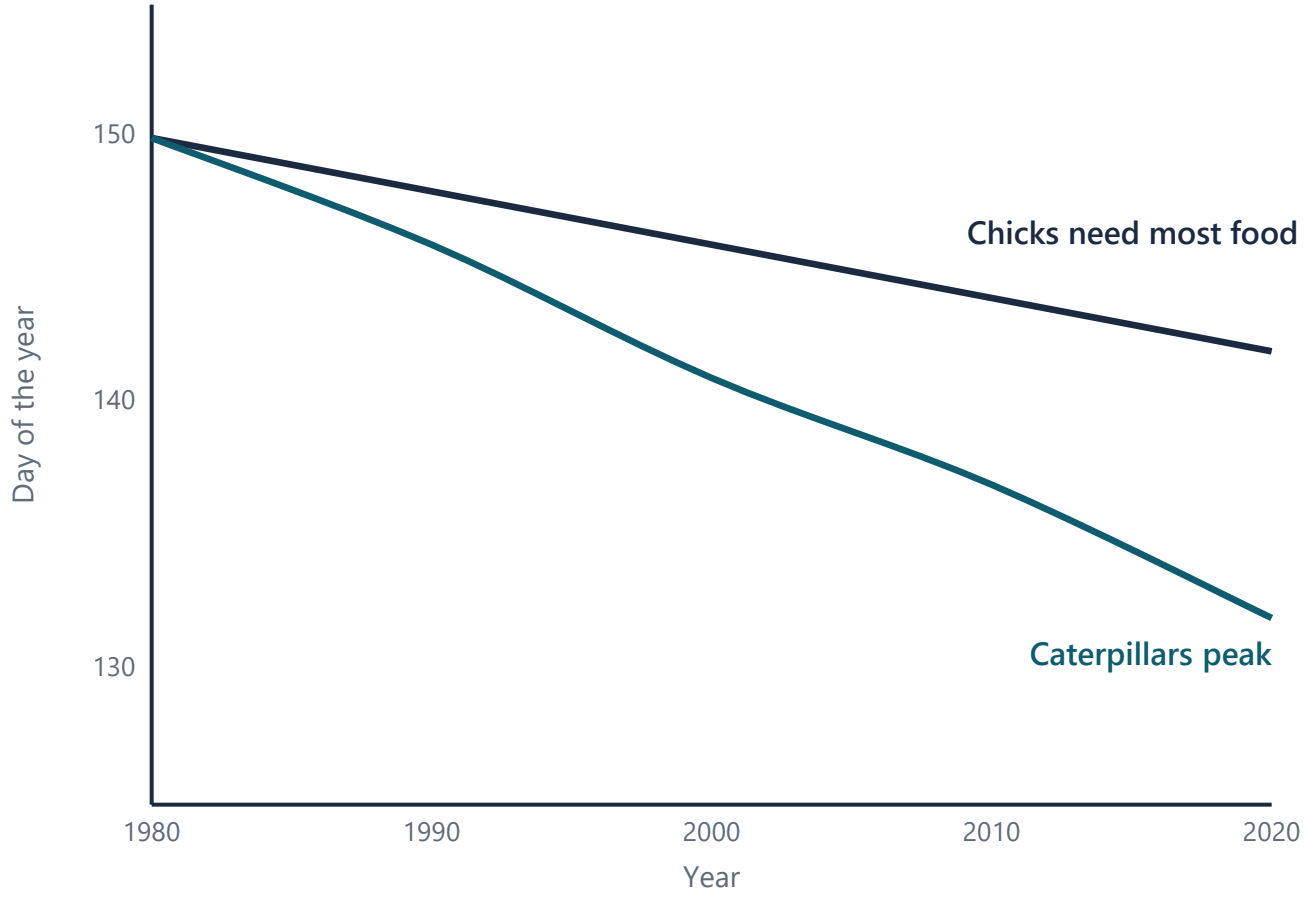
- Arctic mouse-ear and other plants start growing earlier as temperatures rise
- Reindeer migrate in response to day length, so they arrive at about the same date
- Females need the most food when calving
- Plants are now past their most nutritious stage, and fewer calves survive

Think about it

Why does the warming matter more for an animal whose journey is set by day length than for one that responds directly to temperature?

Out of step: great tits and caterpillars

D4.3.10



Schematic trends, not measured values.



Great tit. Dan Davison, CC BY 2.0.

- Caterpillars track spring warmth
- Great tits shift laying less
- Chicks hatch after the food peak

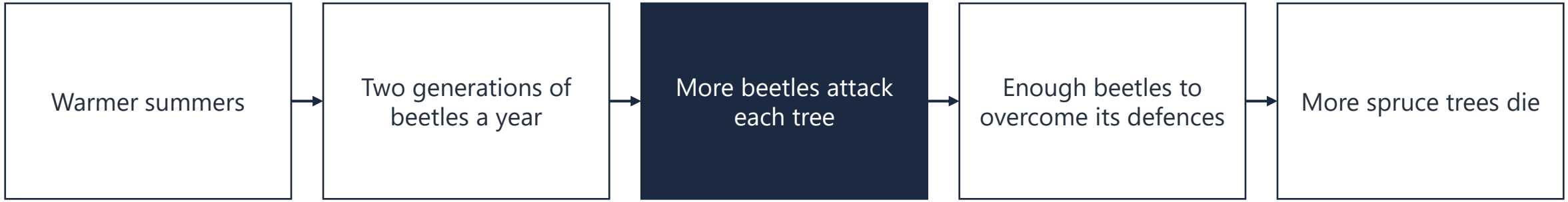
More insect generations

D4.3.11



Spruce bark beetle, Ips typographus. Daniel Adam, Bugwood.org, CC BY 3.0 us.

- Insects develop faster in warmth
- Northern spruce bark beetles have had one generation a year
- Warming allows **two generations** a year
- Beetles can also spread further north



Link

Dead forests can release carbon, adding to warming (D4.3.2–3).

Evolution: tawny owl colour

D4.3.12



Tawny owl. Akumiszcza, CC BY 3.0.

- Tawny owls are pale grey or brown; the colour is heritable
- Grey owls were best hidden in snowy Finnish winters
- Brown owls rose from about 30% to about 50% of the Finnish population

Less snow cover in winter → Pale owls are easier to see against dark ground → Pale owls catch less food and survive less → **Allele for brown feathers rises in frequency**

Key idea

A changed climate changes which variant has the highest adaptive value.

Evolution: bud set in Scots pine

D4.3.12

- **Bud set** is when a tree stops growing and forms protected buds for winter
- Trees differ in the date of bud set, and part of this difference is heritable
- Too early: growing time is wasted. Too late: frost damages new growth
- A longer growing season favours trees that set bud **later**

Prediction

Over generations, the mean date of bud set should move later, as alleles for later bud set rise in frequency.

Limits

Trees have long generations, so evolution may be too slow to keep up with fast climate change. Which species do you think will adapt fastest, and why?

Going further: how scientists track timing

D4.3.9–10

Phenology needs long records. Some come from centuries of diaries; others from today's citizen scientists.

- **Historical records**: cherry blossom dates in Kyoto go back over 1000 years and show recent flowering much earlier than before.
- **Citizen science**: volunteers record first sightings of swallows, frogspawn or leaf burst each year.
- **Satellites**: measure when large areas turn green in spring.
- **Nest boxes**: long studies of great tits record egg-laying dates and caterpillar peaks.

Check-in: phenology and evolution

D4.3.9–14

1. Define phenology.
2. Name the two main cues organisms use to time events.
3. Explain why caribou calves may now miss the peak of plant growth.
4. Explain how a mismatch affects great tit chicks.
5. Explain why some insects now have more generations each year.
6. Give an example of evolution in response to climate change.

How to use this

Answer without your notes, then compare with a partner.

Working with data: caterpillars and chicks

D4.3.11

Day of the year of the caterpillar peak and the great tit hatching peak. These data are invented for practice; the pattern matches long-term studies.

Decade	Caterpillar peak	Hatching peak
1970s	150	148
1990s	142	145
2010s	135	141

1. Calculate the mismatch in each decade.
2. Explain the change in mismatch.
3. Predict the effect on great tit numbers.

Common mistakes to avoid

Not this	But this
All species shift their timing by the same amount.	Species respond differently, which causes mismatches.
Day length changes with climate change.	Day length depends on latitude and date only.
Animals choose to change colour.	Colour morph frequencies change by natural selection.

Check that you can...

1. **Define** phenology and **outline** the cues used to time events
2. **Explain** mismatches caused by climate change, with examples
3. **Explain** why insect life cycles are speeding up
4. **Outline** evolution in response to climate change

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Define phenology.
2. Explain why climate change can put two species out of step.
3. Explain why the proportion of brown tawny owls has increased.

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